

Assignment-II(A)

Function Spaces

(For Submission; Due date: September 1, 2025)

1. Determine (with justification) whether the following statements are true or false:
 - (i) Every continuous function $f \in C[1, 2]$ can be uniformly approximated by a sequence of even polynomials.
 - (ii) Every continuous function $f \in C[1, 2]$ can be uniformly approximated by a sequence of odd polynomials.
 - (iii) Every continuous function $f \in C[0, 1]$ can be uniformly approximated by a sequence of even polynomials.
 - (iv) Every continuous function $f \in C[0, 1]$ can be uniformly approximated by a sequence of odd polynomials.

2. Fix some function $f : \mathbb{R} \rightarrow \mathbb{R}$. Define a sequence of functions $f_n : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f_n(x) = f(nx) \quad (x \in \mathbb{R}).$$

If $\{f_n\}$ forms an equicontinuous family then show that f is constant.

3. Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a sequence of continuous functions such that

$$|f_n(x)| \leq 1 \quad \forall x \in [0, 1] \text{ and } \forall n \in \mathbb{N}.$$

Define a sequence of functions $F_n : [0, 1] \rightarrow \mathbb{R}$ by

$$F_n(x) = \int_0^x f_n(t) dt \quad \forall n \in \mathbb{N} \quad (x \in [0, 1]).$$

Show that $\{F_n\}$ has a subsequence that converges uniformly on $[0, 1]$.

4. Let f be a continuously differentiable function on $[a, b]$. Show that there is a sequence of polynomials $\{P_n\}$ such that $\{P_n\} \rightarrow f$ uniformly and $\{P'_n\} \rightarrow f'$ uniformly on $[a, b]$.
5. Give an example (with justification) of a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ which can not be approximated by any sequence of polynomials on $\mathbb{R}$.

Assignment-II(B)
Function Spaces
(Some Additional Practice Problems)

1. Give an example of an infinite family of non-constant functions that is equicontinuous at a point.
2. Give an example of a metric space X and an algebra in $C(X)$ that fails to separate points and also vanishes at some point.
3. There are three main hypotheses of the Arzèla-Ascoli theorem, namely,
 - (a) The domain of the functions in the sequence $\{f_n\}$ is compact
 - (b) The sequence $\{f_n\}$ is uniformly bounded
 - (c) The sequence $\{f_n\}$ is uniformly equicontinuous.

Show that the theorem is not true by giving a counter-example for each of the following cases

- (i) (a) and (b) hold, but not (c)
- (ii) (b) and (c) hold, but not (a)
- (iii) (a) and (c) hold, but not (b).